FF-P-00416F(GSA-FSS)
November 15, 1971
INTERIM REVISION OF
Fed. Spec. FF-P-416e
December 5, 1963

INTERIM FEDERAL SPECIFICATION

PIN, SAFETY

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification FF-P-416e, dated December 5, 1963

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers general purpose corrosion resisting safety pins of various sizes with or without guard.

1.2 Classification. The safety pins shall be of one type and the following classes and sizes (see 3.4 and table I), as specified (see 6.2):

Class 1 - With guard.
Class 2 - Without guard.

Size 1 - Small, length 1-1/16 inches.
Size 2 - Medium, length 1-1/2 inches.
Size 3 - Large, length 2 inches.
Size 6 - Extra large, length 3 inches.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-B-613 - Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet and Strip).
QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.
QQ-W-321 - Wire, Copper Alloy.
QQ-W-423 - Wire, Steel, Corrosion-Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.
PPP-B-676 - Boxes, Set-Up.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Int. Fed. Test Method Std. No. 00304 - Metals, Chemical Test Methods (For Identification).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

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(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.
MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The safety pins shall be manufactured of copper zinc alloy (brass), or corrosion-resisting steel (stainless steel), at the option of the manufacturer.

3.1.1 Copper zinc alloy wire. The wire shall be spring tempered wire, alloy number 260, 270, or 274 of QQ-W-321.

3.1.2 Copper zinc alloy strip. The strip shall be annealed strip, alloy number 240, 260, or 268 of QQ-B-613.

3.1.3 Corrosion-resisting steel wire. The steel wire shall be spring tempered wire, composition 302, 304, 305, 316, or 347 of QQ-W-423.

3.1.4 Corrosion-resisting steel strip. The steel strip shall be annealed steel, class 302, 304, 305, 316, 321, 347, or 430 of QQ-S-766.

3.2 Design and construction. The safety pins shall be made of material as specified in 3.1. (See 4.4.) The design and construction shall conform to standard commercial practices.

3.2.1 Spring. The spring shall be formed by one and one-half turns of the wire.

3.2.2 Point. The free end of the pin shall gradually taper to a sharp point.

3.2.3 Cap. The cap shall be designed and constructed so as to permit easy closure and opening from either side of cap, but shall prevent accidental opening. In closed position, point of pin shall be at least halfway into cap.

3.2.4 Guard (class 1 only). The guard shall be securely clamped around approximately one-third to one-half of the circumference of the spring.

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3.3 Finish. All surfaces of the copper zinc alloy safety pins shall be uniformly plated with nickel of a thickness used in standard commercial practices. (See 4.4.1.3.) Corrosion resisting steel pins shall have a No. 2 D bright finish, or a No. 4 general purpose polished finish.

3.3.1 Resistance to corrosion (copper zinc alloy pins only). When tested for resistance to corrosion as specified in 4.4.1.2, the total surface of any pins showing corrosion shall not exceed 2 percent.

3.4 Dimensions. The safety pin shall conform to the sizes and measurements shown in table I (see 1.2).

TABLE I. Dimensions

Commercial size	Length of safety pins. (Tolerance ± 1/16 inch)	Diameter of finished wire. (Tolerance ± 0.001 inch)		Thickness of metal strip for cap (Tolerance ± 0.001 inch)		Minimum thickness of strip metal for guard (class 1 only)
	inches	inch		inch		inch
		Brass	Stainless Steel	Brass	Stainless Steel	
1 - Small	1-1/16	0.033	0.030	0.010	0.009	0.006
2 - Medium	1-1/2	.041	.036	.010	.009	.006
3 - Large	2	.050	.043	.012	.009	.006
6 - Extra large	3	.071	.045	.016	.009	.006

3.5 Identification marking. Each box of safety pins shall be legibly marked with the name or trademark of the manufacturer.

3.6 Workmanship. The finished safety pins shall be clean and uniform. The occurrence of defects shall not exceed the acceptable quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection and testing of base material. In accordance with 4.1, the supplier is responsible for insuring that base material is manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications, standards and other documents.

4.2 Sampling.

4.2.1 Size and lot. All safety pins of the same class, and size presented for inspection at one time shall be considered a lot for purpose of examination and test.

4.2.2 Sampling and inspection for acceptance. Sampling and inspection for acceptance shall be performed in accordance with MIL-STD-105, except where otherwise specified.

4.2.3 Sampling for visual inspection. Random sample boxes of safety pins shall be taken from each lot in accordance with MIL-STD-105. A sample unit shall be one box of safety pins. The pins shall be examined for defects listed in table II. The Inspection Level shall be S-4, and the Acceptable Quality Level (AQL) 4 percent defective.

4.2.4 Sampling for preparation for delivery. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The sample shall be examined for defects listed in table III. The lot size shall be the number of shipping containers in the end-item inspection lot. The Inspection Level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

4.2.5 Sampling for tests. For each test, random sample boxes of safety pins shall be taken from each lot in accordance with MIL-STD-105. A sample unit shall be one box, except that five pins, taken at random, shall be tested as specified in 4.4. The Inspection Level shall be S-2, and the AQL 2.5 percent defective.

4.3 Examination.

4.3.1 Visual examination of end-item. Samples taken in accordance with 4.2.3 shall be examined for defects listed in Table II. Any safety pin in the sample having one or more defects shall be rejected. If the number of defective units in the sample exceeds the acceptance number for that sample, the entire lot represented by the sample shall be rejected.

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TABLE II. Visual examination

Examine	Defects
Material	Material not as specified (see 3.1).
Class and size	Class and size not as specified (see 1.2, 3.4, and table I).
Design and construction	Design and construction not as specified (see 3.2).
Finish	Finish not as specified (see 3.3).
Identification marking	Boxes of safety pins not marked as specified (see 3.5).
Workmanship	Workmanship not in accordance with 3.6).

4.3.2 Examination of preparation for delivery. Samples selected in accordance with 4.2.4 shall be examined for defects listed in table III to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with.

TABLE III. Examination of preparation for delivery

Examine	Defects
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; or improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.3.2.1 Examination for closure and reinforcement of containers. Shipping container shall be examined for defects in closure and reinforcement in accordance with the appendix of the applicable shipping container specification.

4.4 Tests.

4.4.1 End item tests. Samples selected in accordance with 4.2.5 shall be subjected to the following tests. Any safety pin in the sample failing the test shall be considered defective. If the number of defective pins exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4.1.1 Magnetic properties (copper zinc alloy only). The sample pins shall be brought in contact with a permanent magnet to determine whether carbon steel base material has been employed in the manufacture of the safety pins. Carbon steel pins shall be rejected.

4.4.1.2 Resistant to corrosion (copper zinc alloy only). The sample pins shall be boiled in distilled water in a glass beaker for 30 minutes and allowed to remain submerged for 24 hours. The surface of the pins shall then be examined for corrosion (see 3.3.1).

4.4.1.3 Identification for presence of nickel coating (copper zinc alloy only). To determine presence of nickel coating, the safety pins shall be tested in accordance with Test Method 108 in Int. Fed. Test Method Standard 00304.

4.4.1.4 Identification of corrosion-resisting steel. To identify corrosion-resisting steel, the safety pins shall be tested in accordance with test method 107 in Int. Fed. Test Method Standard 00304.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified, pins in unit quantities of the same size, shall be packaged in boxes conforming to PPP-B-566, type and style optional, or PPP-B-676, type and style optional. The number of pins per box shall be as shown in table IV. Each box shall be waterproofed in accordance with MIL-P-116, method IC-2.

5.1.2 Level B. Pins shall be packaged in boxes as specified in 5.1.1 except that waterproofing shall not be required. For size 2, twelve pins shall be mounted on a card of suitable size and weight and of such design that each pin will be securely held in position, or twelve pins shall be packaged in a heat-sealed polyethylene bag, minimum 0.002 inch thickness.

5.1.3 Level C. Safety pins shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

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5.1.4 Intermediate packaging. Twelve unit packages of pins shall be intermediate packaged in a box conforming to PPP-B-566 or PPP-B-676, type and style optional. Sealing and closure shall be in accordance with the appendix to the box specification.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The unit and intermediate containers packaged as specified in 5.1, shall be packed in quantities in accordance with table IV, in any of the following containers at the option of the contractor:

Specification	Type or Class
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class domestic
PPP-B-640	Class 2

The gross weight of wood boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall be subject to the limitation of the box specification. Each shipping container shall be provided with a sealed case liner conforming to MIL-L-10547. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Intermediate packages of pins shall be overpacked in quantities specified in table IV in a box conforming to PPP-B-636, class domestic. The container shall be closed and sealed in accordance with the appendix to the box specification.

5.2.3 Level C. Safety pins in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable

5.3 Marking. In addition to any special marking required in the procurement documents, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. 123 for civil agencies or MIL-STD-129 for Military agencies, as applicable (see 6.2).

TABLE IV. Packaging and packing

Commercial size	Length (inches)	Pins per unit package	Unit packages per intermediate container	Intermediate containers per shipping container
1	1-1/16 (small)	96	12	24
2	1-1/2 (medium)	12 or 72 as specified (see 6.2)	12	12
3	2 (large)	48	12	12
6	3 (extra large)	24	12	12

6. NOTES

6.2 Intended use. The safety pins are intended for general use. Class 1 pins, with guard on the coil, are intended to reduce the tendency of thread of fabric becoming caught in the coil.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class and size required (see 1.2 and 3.4).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) Whether special marking is required (see 5.3).
(e) Quantities required (see 5.1.1, 5.1.2 and 5.2.3).
(f) Marking document required (see 5.3).

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